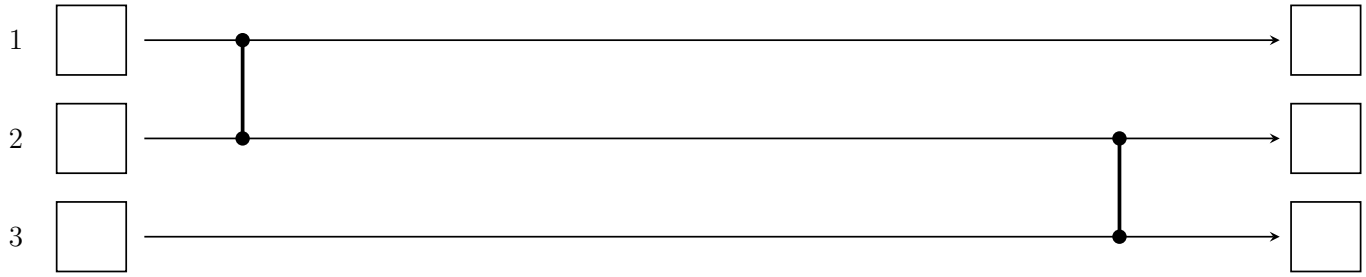
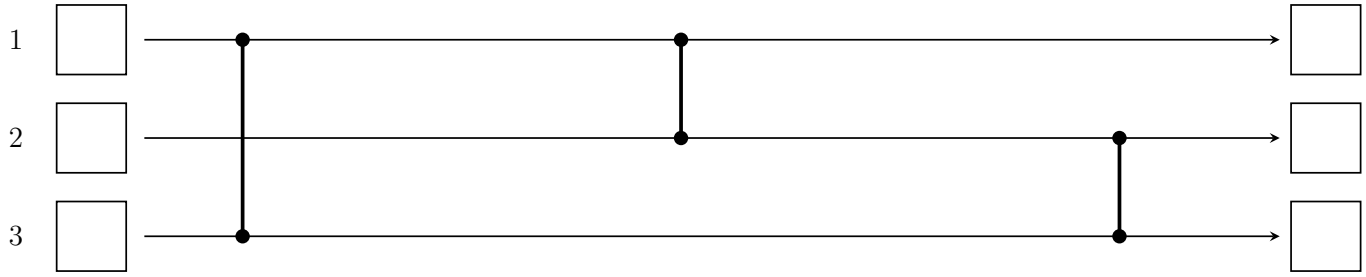
Move from left to right. At each bar, compare only the two lanes with black dots: the smaller value goes to the upper lane and the larger to the lower lane. Ties may stay put. Keep every bar fixed for every start; move cards only at bars. A machine sorts when the final values never get smaller from lane 1 down.

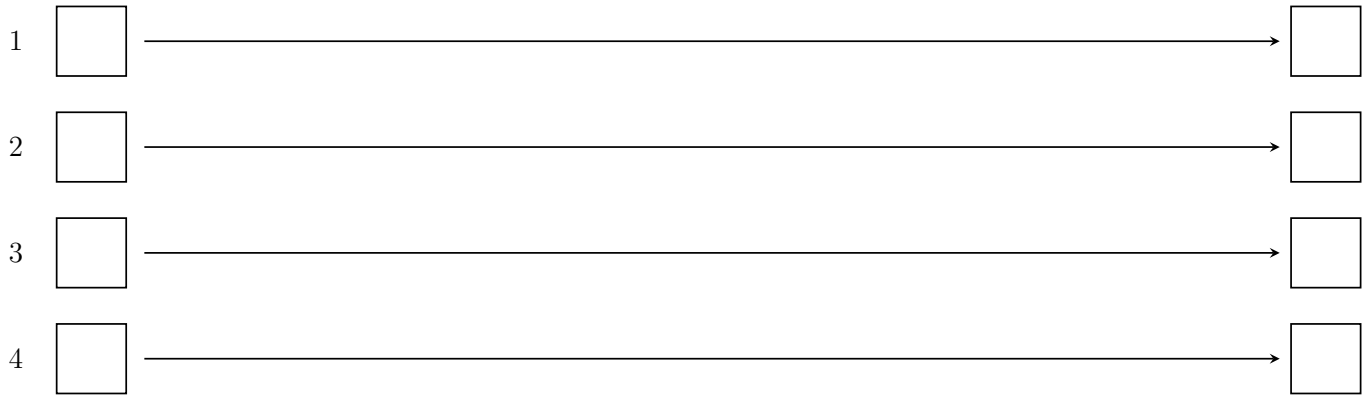
Problem 1: For each machine, decide whether it sorts every order of 1, 2, 3. Give a failing start or a complete test record. Decide whether either machine can fail with 1, 1, 3 or 1, 3, 3.



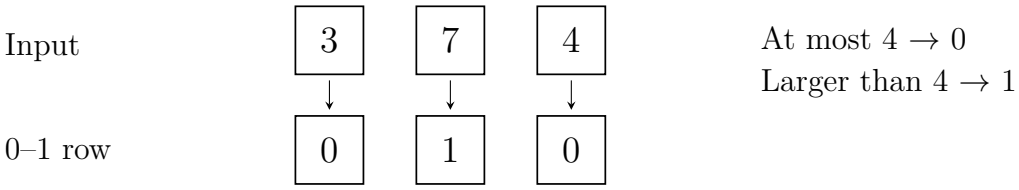


Problem 2: Build a four-lane machine that sorts every order of 1, 2, 3, 4. Use as few bars as you can.





Example: choose $t = 4$. Keep each card in its position.



Problem 3: Choose a number t. Replace a value by 0 when it is at most t, and by 1 when it is larger than t. For each start, find every different row of 0s and 1s that choosing t can produce.

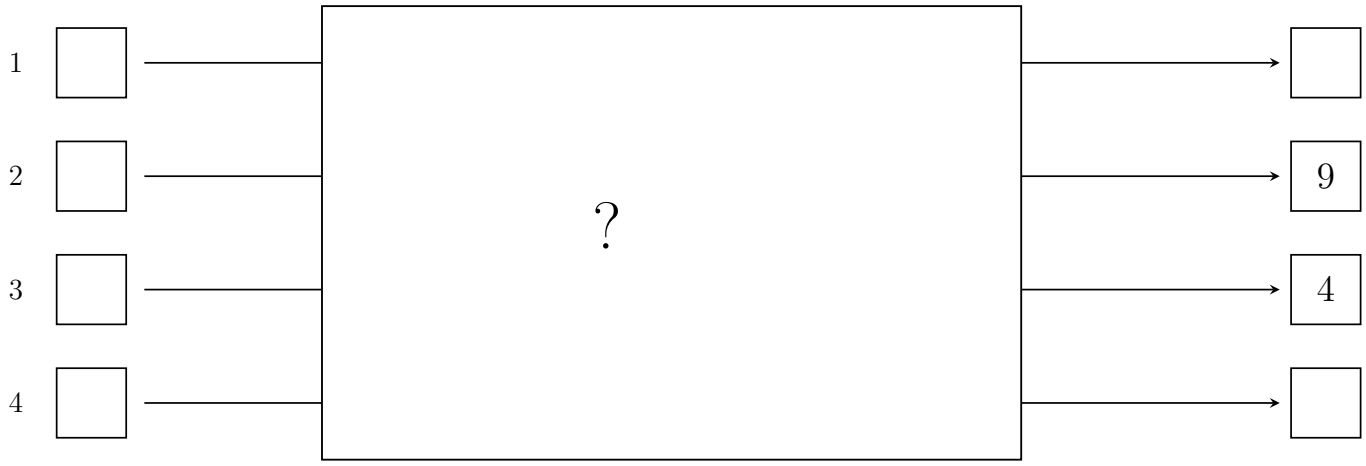
-2	8	5	8

6	2	9	4

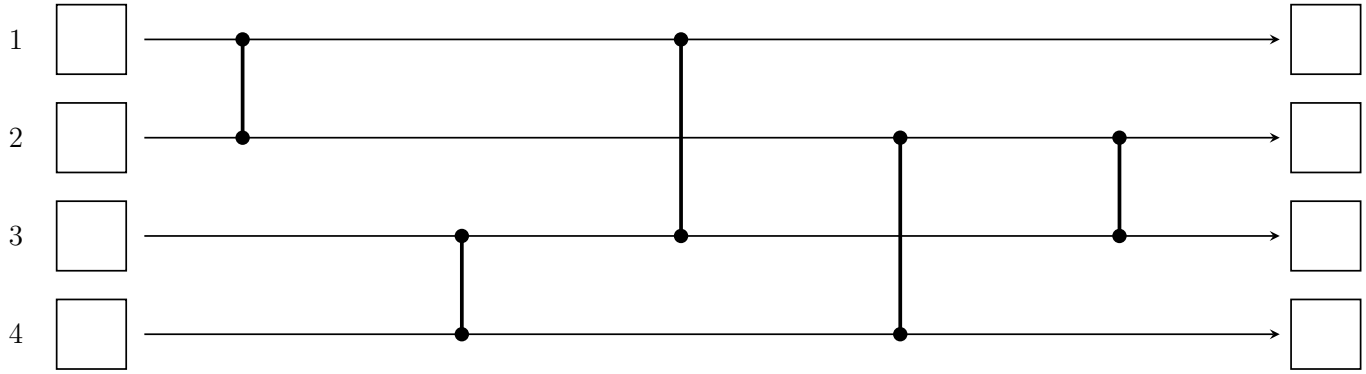
4	4	4	4

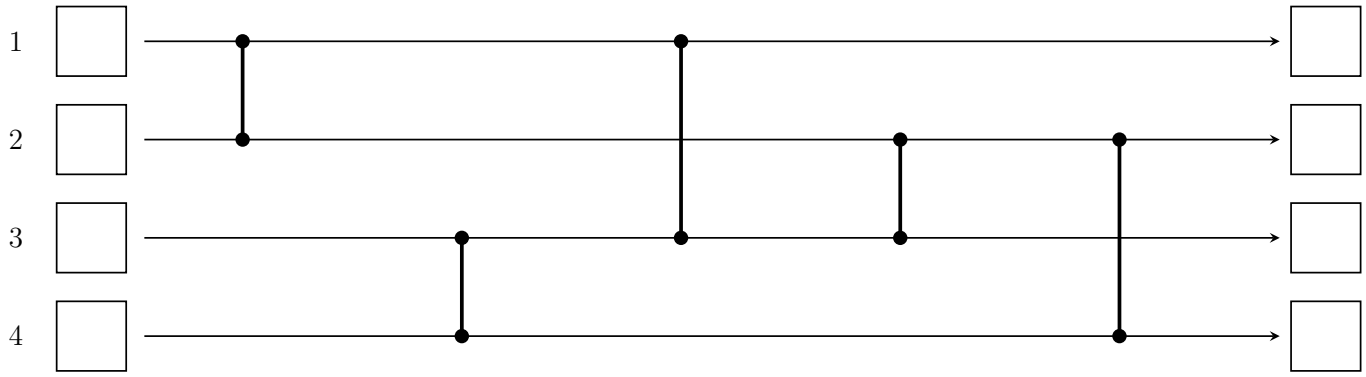
Can replacing values before a bar ever give different final 0s and 1s from replacing them after the bar? Explain for any two values and any t.

Problem 4: A fixed four-lane machine sorts every start made of 0s and 1s. Could some other start finish with 9 in lane 2 and 4 in lane 3? Explain. Decide whether sorting every 0–1 start guarantees sorting every start made of numbers, including repeats.

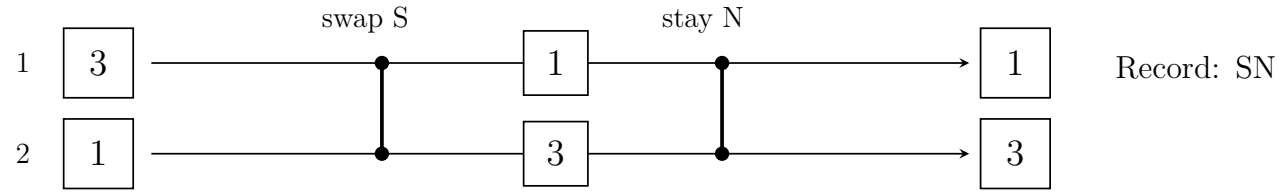


Problem 5: Decide which machine sorts every start made of numbers. Give an explanation covering all inputs, including repeated values.

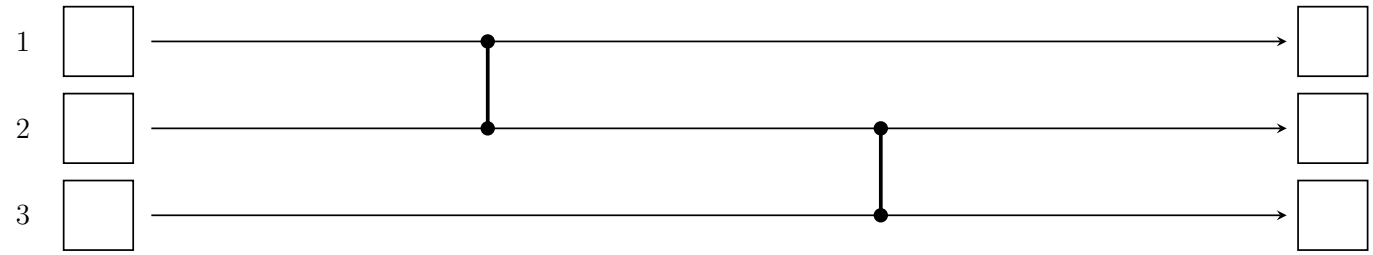


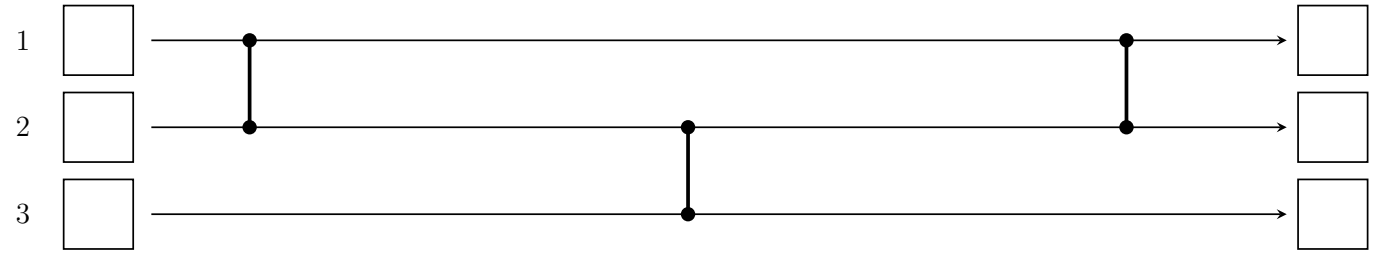


Example: compare the same two lanes twice.



Problem 6: Write S when two cards swap at a bar and N when they stay put, in order from left to right. For each machine, find every record made by orders of 1, 2, 3, and find every start that makes each record.





Could a fixed machine sort two different starting orders of distinct cards while making the same swap/no-swap record? Explain.

Problem 7: Find the fewest bars a three-lane sorting machine can have and the fewest bars a four-lane sorting machine can have. Give a working machine for each and explain why fewer bars cannot work.

